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PROFESSIONAL SUMMARY

Backend-focused Computer Science student with hands-on production experience designing and shipping **RESTful APIs, real-time systems, and caching layers** using **Node.js/Express, FastAPI, MongoDB, PostgreSQL, and Redis**. Experience deploying and monitoring services in production (**Render, Docker, Sentry**) across **6 merged pull requests** as a developer intern and **5+ hackathons**, backed by coursework in operating systems, databases, networking, and distributed system design.

EDUCATION

Canara Engineering College, Mangalore

CGPA: 8.26

KSEAB (SSLC/PUC)

B.E in Computer Science & Engineering

Expected Graduation: 2026

SSLC: 88.96% — PUC: 92.96%

EXPERIENCE

Orbit — Intern, Technical Team Member

Feb 2026 – Present

- Built and maintained **REST APIs** in **Node.js/Express** for a location-based social discovery platform, working within a monorepo alongside a React Native/Expo client.
- Designed a **Redis-backed geocoding cache layer** in front of Nominatim/Photon geocoding services, defining **TTL strategies** and cache-key conventions to cut redundant external API calls and reduce lookup latency.
- Engineered real-time chat infrastructure using **WebSockets** and geolocation-based event discovery services powering the platform's core features.
- Deployed and maintained the backend in production on **Render** with **Docker**-based environment consistency and **Sentry** error monitoring; diagnosed and resolved reliability issues including an overly strict OTP rate limiter, a blocking SMTP call, and a WebSocket-related memory leak (OOM) flagged during pre-deployment review.

ACHIEVEMENTS & LEADERSHIP

Core Unleashed Hackathon — Team Lead, “Digital Sentinels”

Apr 2025

- Ranked among the **Top 10 teams** out of **72 teams** and pitched our solution to the final judging panel after advancing through multiple evaluation rounds.
- Led a team to build PII Shield, a privacy-first PII redaction tool, in a **36-hour hackathon**, designing a **stateless backend** with a deterministic finding-ID system and running **OCR (Tesseract)** entirely locally to guarantee zero data retention.
- Implemented a **3-tier bounding-box fallback** (exact character → line-level → sidebar-only) to keep redaction accurate despite imperfect OCR spatial mapping.

Feature Development & Deployment

Ongoing

- Merged **5+ pull requests** into Orbit's production codebase, shipping backend fixes and new API functionality.
- Implemented **event creation with feed-level visibility**, enabling public events to surface across all users' feeds.
- Built a **location-picker** that reverse-geocodes a user-dropped map pin into an address.

PROJECTS

HVSapp — Healthcare Handoff Validation Platform

FastAPI, PostgreSQL, Celery, Redis

- Built the backend for a clinical handoff and medication-tracking system using **FastAPI**, decoupling an ASR transcription pipeline into async **Celery** task queues backed by **Redis** to avoid blocking real-time request handling.
- Implemented **PHI field encryption** and medication safety tracking via **Alembic** migrations on **PostgreSQL**, structuring the schema around clear domain boundaries (auth, medication, patients, transcription).
- Applied targeted database indexing and async task decoupling to improve API responsiveness under concurrent clinical workflows.

GraphSentinel — Graph Neural Network Cyber Defense System

Python, PyTorch Geometric, GraphSAGE

- Designed and trained a GraphSAGE-based intrusion detection model on CICIDS-2017, engineering a **7-feature behavioral signal set** from raw traffic data and resolving severe class imbalance (128K vs. 1,956 samples) with hybrid under-sampling and SMOTE.
- Achieved **99.57% accuracy** and **0.9975 AUC-ROC**, with inference optimized to **~120ms per graph** to fit within a 5-second real-time detection budget.

TECHNICAL SKILLS

- **Backend & APIs:** Node.js, Express, FastAPI, Flask, RESTful API design, WebSockets, Celery (async task queues)
- **Databases & Caching:** MongoDB, PostgreSQL, Redis (caching & TTL design), Alembic migrations
- **Cloud & DevOps:** AWS (EC2, S3, VPC), Docker, Render, Git/GitHub, Sentry, CI/CD basics, Linux Administration
- **Languages:** Java, Python, JavaScript, SQL
- **Frontend & Mobile:** React.js, React Native, Redux, Tailwind CSS
- **ML & Data:** PyTorch, PyTorch Geometric (GraphSAGE), Scikit-learn, OpenCV, MediaPipe